LESSON PLAN

GUN SYSTEM (35)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Warhead systems requirements and types
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Warhead systems requirements and types
5. Specific Objectives	35.1.1.1 Trainee will be able to explain role of underwater weapons

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of underwater weapon warhead diagrams; Safety: Do not handle live components; Instructor supervision: Monitor diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a warhead in a torpedo?
(b) How does the underwater environment affect warhead design?
(c) Name one key requirement for warhead safety in storage.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General requirements for warhead systems • Safety mechanisms and arming procedures • Detonation methods and initiation systems • Compatibility with torpedo body structures • Environmental sealing and pressure resistance	L & Ds	PPT, IB	15 min
(b)	• High explosive types used in naval torpedoes • Fuzing systems for underwater targets • Warhead sizing and weight distribution • Thermal and shock protection features • Maintenance and inspection criteria for warheads	L & Ds	PPT, PDF	20 min
(c)	• Comparison of contact vs. magnetic fuzing • Role of warheads in anti-submarine warfare • Impact of depth on warhead effectiveness • Future trends in warhead technology • Integration with control and power systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between contact and magnetic fuzing in underwater weapons.
(b) Describe how warhead requirements change based on target depth.

12. Home Assignment and Follow Up	Prepare a brief note on how the warhead system interfaces with the propulsion and control systems discussed in LP 06.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Propulsion system used in torpedo, power supply and control systems	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Petty Officer, P No. 12345	Senior Instructor, Chief Petty Officer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min